# HDOC Day-55

## First Negative Integer in Every Sliding Window

## Aim

To write a C program using pointers and a nested-loop approach to find the first negative integer in every contiguous subarray (sliding window) of size k.

## Problem Statement

- Read the size and elements of a variable-length integer array, arr[], and an integer k.
- Pass the array to first_negative() using its address (pointer).
- For every contiguous window of size k, find the first negative integer from left to right.
- Display 0 when a window has no negative integer.
- Separate the results by spaces.
- Display -1 when k is invalid (k <= 0 or k > size).
- Use a nested-loop approach.

## Algorithm

1. Read size and the array elements.
2. Read k and call first_negative(arr, size, k).
3. If k is invalid, print -1 and return.
4. Use the outer loop to move the starting position of each window from 0 to size - k.
5. For each window, use an inner loop to check elements from left to right.
6. Print the first negative value found; if none is found, print 0.

## C Program

```c
#include <stdio.h>

void first_negative(int *arr, int size, int k)
{
    int i, j, found;

    if (k <= 0 || k > size)
    {
        printf("-1");
        return;
    }

    for (i = 0; i <= size - k; i++)
    {
        found = 0;

        for (j = i; j < i + k; j++)
        {
            if (*(arr + j) < 0)
            {
                printf("%d", *(arr + j));
                found = 1;
                break;
            }
        }

        if (found == 0)
            printf("0");

        if (i < size - k)
            printf(" ");
    }
}

int main()
{
    int size, i, k;

    printf("Enter the size of array: ");
    scanf("%d", &size);

    int arr[size];

    printf("Enter the array elements: ");
    for (i = 0; i < size; i++)
        scanf("%d", &arr[i]);

    printf("Enter k: ");
    scanf("%d", &k);

    printf("Output: ");
    first_negative(arr, size, k);

    return 0;
}
```

## Test Cases

| Test | Array | k | Expected Output |
|---|---|---|---|
| 1 | 12 -1 -7 8 -15 30 16 28 | 3 | -1 -1 -7 -15 -15 0 |
| 2 | 1 2 -3 4 5 -6 | 2 | 0 -3 -3 0 -6 |
| 3 | 1 2 3 4 | 2 | 0 0 0 |
| 4 | 1 2 3 | 4 | -1 |

## Sample Outputs

```
Test Case 1
Enter the size of array: 8
Enter the array elements: 12 -1 -7 8 -15 30 16 28
Enter k: 3
Output: -1 -1 -7 -15 -15 0
```

```
Test Case 2
Enter the size of array: 6
Enter the array elements: 1 2 -3 4 5 -6
Enter k: 2
Output: 0 -3 -3 0 -6
```

```
Test Case 3
Enter the size of array: 4
Enter the array elements: 1 2 3 4
Enter k: 2
Output: 0 0 0
```

```
Test Case 4
Enter the size of array: 3
Enter the array elements: 1 2 3
Enter k: 4
Output: -1
```

## Result

The program successfully finds and displays the first negative integer in each sliding window of size k using pointers and nested loops. It displays 0 when no negative integer is present and -1 for an invalid value of k.